

ReguLens

Drug Safety Signal Detector & Regulatory Submission Readiness Checker

Team BOB404

TRACK: AI | ML

THE PROBLEM

Evidence review is manual, scattered, and hard to trace

Pharmacovigilance and regulatory-affairs teams assess safety reports and CTD dossiers through spreadsheets, keyword search, and separate source files — before they can decide what needs investigation or whether a submission is ready.



Who is affected

Pharmacovigilance analysts investigating adverse-event patterns, and regulatory-affairs teams preparing or reviewing CTD submissions.



Why it matters

Slow review delays signal investigation; weak traceability means a reviewer can't quickly verify the source evidence behind a conclusion.



Why keyword search falls short

Related CTD evidence uses different wording than the requirement text, and must still be judged in the correct module and section context.

TWO REVIEW WORKFLOWS. ONE BLIND SPOT.

Safety-signal review

FAERS adverse-event data reviewed manually for potential signals.

Dossier-readiness review

CTD PDFs checked against ICH expectations, evidence by evidence.

Neither gives reviewers one traceable path from result to source evidence.

One workspace, two traceable review modes

ReguLens turns raw safety data and CTD dossiers into reviewer-ready results while preserving a path back to the source evidence behind every conclusion.

MODE 1

Safety-Signal Analysis

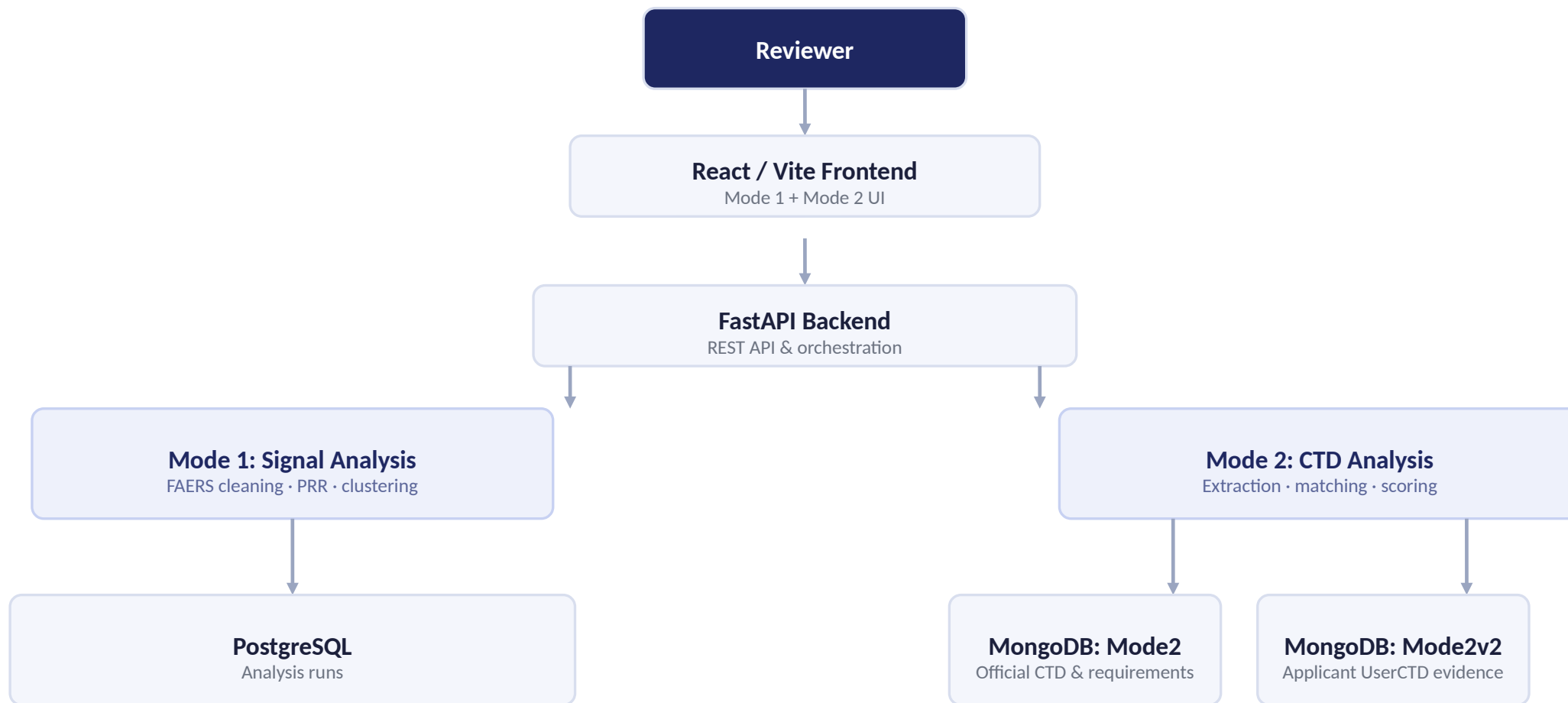
- 1 Upload FAERS CSV data
- 2 Clean records & calculate PRR
- 3 Cluster related adverse events
- 4 Rank & persist candidate signals

MODE 2

CTD Readiness Analysis

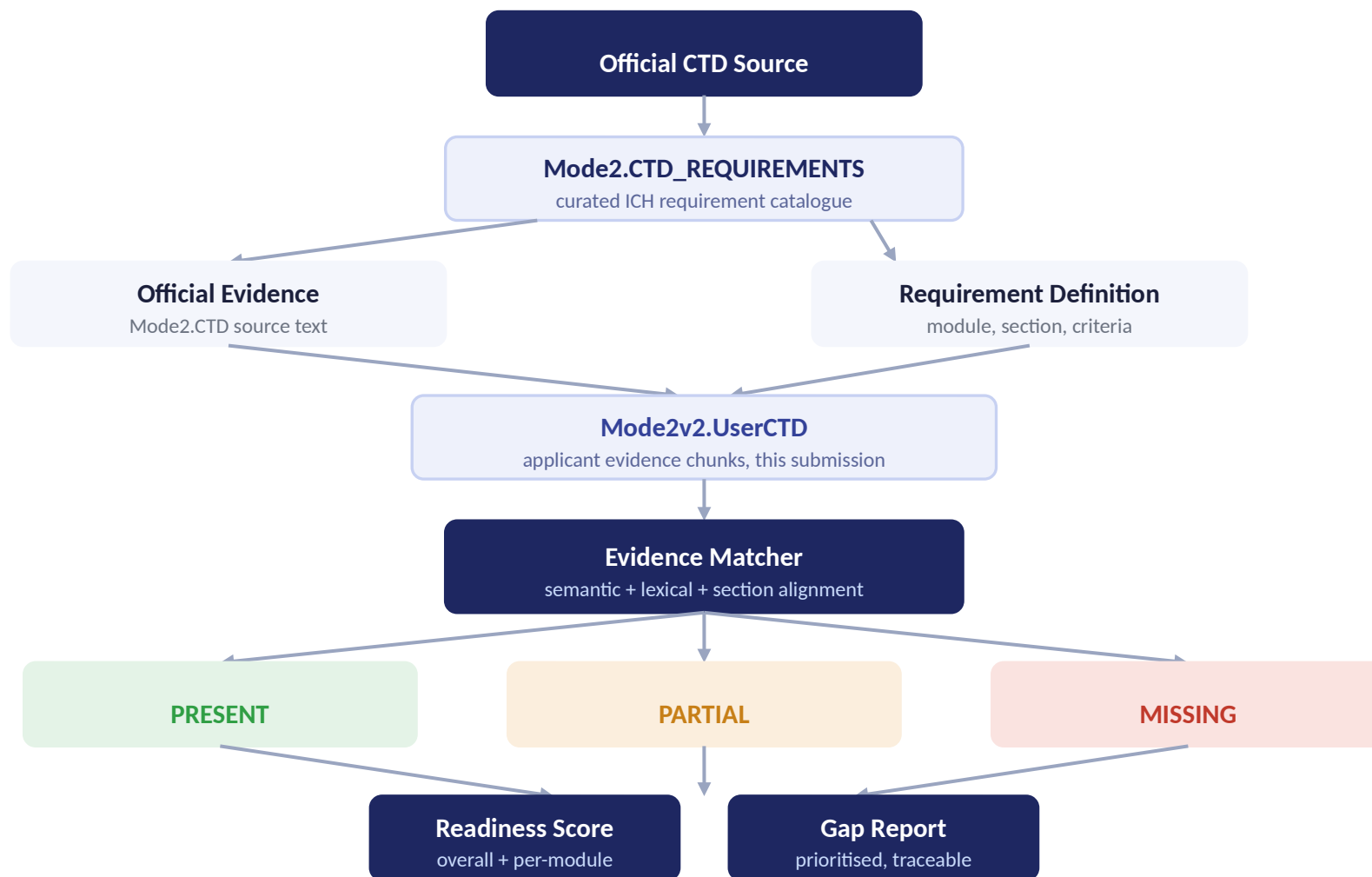
- 1 Upload official + applicant PDFs
- 2 Extract section-aware evidence chunks
- 3 Match evidence to ICH requirements
- 4 Score modules & report gaps

How the system works, end to end



Docling · SentenceTransformers · PyMuPDF support extraction and matching inside Mode 2.

Mode 2: how a CTD requirement gets its verdict



Built and enriched with IBM tools



IBM Bob

AI-assisted development environment used to build and iterate on ReguLens — including the Mode 2 evidence-analysis improvements.



IBM Docling

Structural-enrichment layer in Mode 2. Supplements PyMuPDF text extraction on CTD PDFs while the pipeline retains detected headings and page provenance.

What success looks like for a reviewer

1

One workspace

Safety-signal review and CTD readiness review in a single, traceable workflow instead of separate spreadsheets and files.

2

Evidence-linked results

Every PRR value, cluster, or requirement verdict keeps its source dataset record or dossier page and section.

3

Section-aware matching

Hybrid ranking — embeddings, lexical overlap, CTD-section alignment — so differently-worded evidence is still found.

4

Resilient by design

A deterministic retrieval fallback keeps reports useful even if a remote vector index is temporarily unavailable.

Honest scope: results support evidence review and prioritisation — they are not a substitute for a qualified regulatory decision.

Team: BOB404 Track: AI | ML



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TEAM LEAD & BACKEND

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Built the main server and connected the databases.

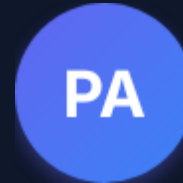


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AI & DATA

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Worked on reading PDFs and integrating AI tools.



Priya Aghera

FRONTEND DEVELOPER

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Designed and built the user screens.



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FULL STACK & TESTING

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Helped process data and tested the app.